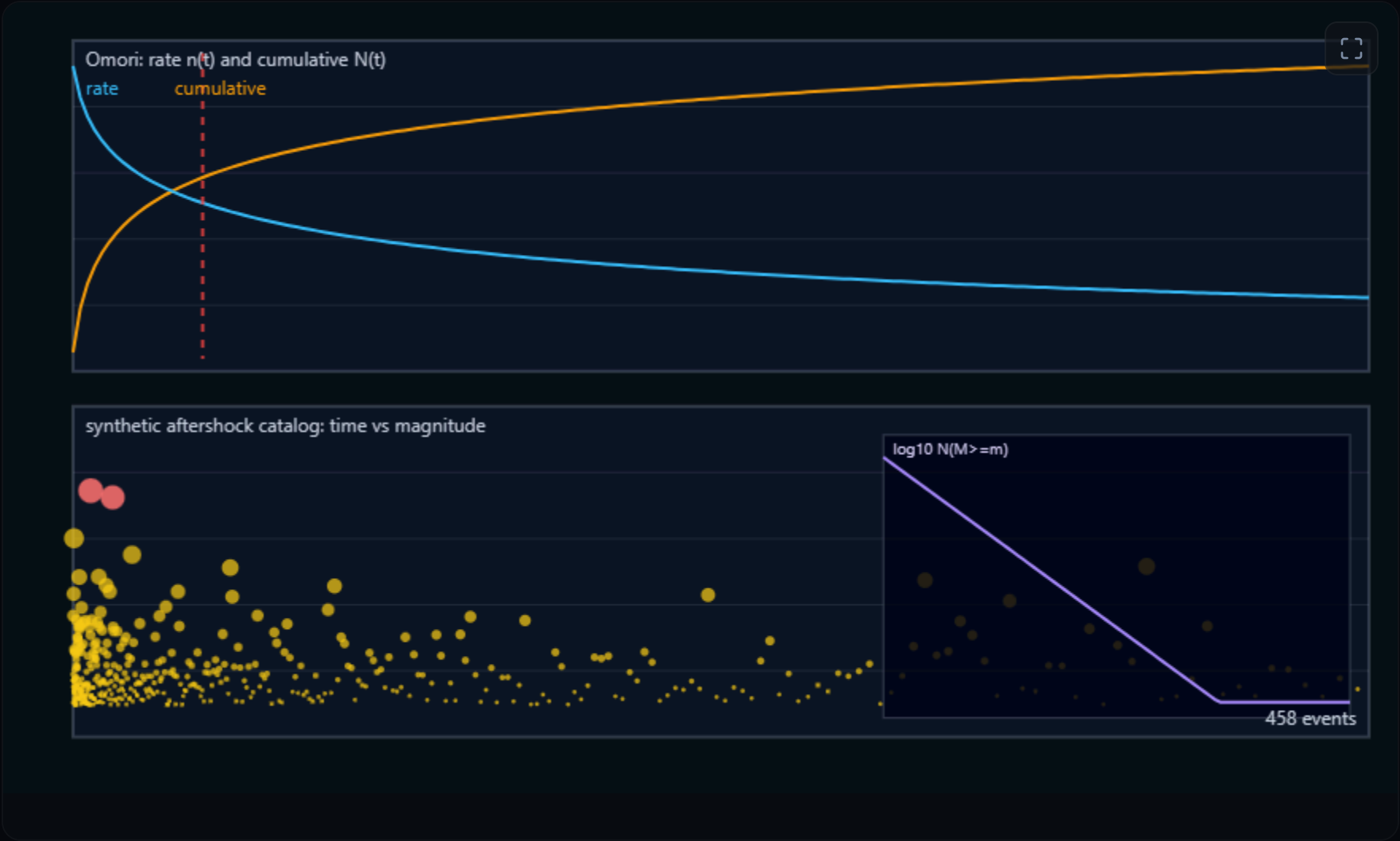


Earthquake Aftershocks: Omori + Gutenberg-Richter

Modified Omori aftershock decay $n(t)=K/(t+c)^p$ combined with Gutenberg-Richter magnitude-frequency curves, b-value, and a synthetic catalog.

Real operational aftershock forecasts include spatial kernels, mainshock magnitude scaling, catalog incompleteness, and uncertainty. This page isolates the two classic empirical laws so their slopes and time scales are easy to read.

LIVE GRAPHS



OMORI LAW

Productivity K

95

Early-time offset c

0.18 d

Decay exponent p

1.08

Catalog window

30 d

Probe day

3 d

GUTENBERG-RICHTER

b-value

0.95

Completeness M_c

2.5 M

Display M_{max}

6.4 M

Omori controls how aftershock rate decays with time; Gutenberg-Richter controls how many small events occur relative to large ones. The catalog is deterministic for display, not a forecast.

No. It shows expected rates and frequency distributions from simple empirical laws. Individual aftershocks are stochastic and operational models require uncertainty and spatial information.

What does a larger b-value mean?

A larger b-value steepens the magnitude-frequency curve, meaning relatively more small events and fewer large events above the completeness magnitude.

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